

Differentiability:

Derivative of a funcⁿ at a point:

Let $f: [a, b] \rightarrow \mathbb{R}$ be a funcⁿ and $c \in (a, b)$ then f is said to be derivable or differentiable at c if

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

exists.

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{f(c) - f(c-h)}{h}$$

Left Hand Derivative

$f: [a, b] \rightarrow \mathbb{R}$, $c \in (a, b)$

$$\text{If } \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{f(c-h) - f(c)}{-h} \text{ exists}$$

then this limit is called L.H.D of f at c and denoted by $f'(c-0)$ or $f'(c-)$ or $Lf'(c)$.

Right Hand derivative

$$\text{If } \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} \quad \text{or} \quad \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} \text{ exists}$$

then this limit is called R.H.D of f at c and denoted by $f'(c+0)$ or $f'(c+)$ or $Rf'(c)$.

$$\text{If } Lf'(c) = Rf'(c)$$

then we say $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ exists, means it is a finite real number.

~~Differ~~

Derivability in an Interval

$f: [a, b] \rightarrow \mathbb{R}$ is said to be derivable in (a, b) if $f'(c)$ exists $\forall c \in (a, b)$.

i.e. $f: [a, b] \rightarrow \mathbb{R}$ is said to be derivable ~~E~~ in $[a, b]$ if

- (i) $f'(c)$ exists $\forall x \in (a, b)$
- (ii) $Rf'(a)$ exists
- (iii) $Lf'(b)$ exists

Derivability and Continuity

Result: If f is derivable at a point 'a' then f is c'ty at point 'a'

but converse is not true.

continuity $\nRightarrow$ derivability

ex $f(x) = |x|$ at $x=0$

$$f(0) = |0| = 0$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} |0-h| = \lim_{h \rightarrow 0} h = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} |0+h| = \lim_{h \rightarrow 0} h = 0$$

$\Rightarrow f(x)$ is c'ty at $x=0$

$$Lf'(0) = \lim_{x \rightarrow 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1$$

$$Rf'(0) = \lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$$

$Lf'(0) \neq Rf'(0)$ so f is not diff. at $x=0$

Q find derivative of $f(x)$ at $x=0$

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Soln

$$Lf'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x} - 0}{x}$$

$$= \lim_{h \rightarrow 0} \frac{(0-h)^2 \sin(\frac{1}{0-h}) - 0}{0-h} \quad \text{put } x = 0-h, h > 0$$

$$= \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h}}{h}$$

$$= \lim_{h \rightarrow 0} h \sin \frac{1}{h}$$

$$= 0 \times \text{a finite quantity b/w } -1 \text{ \& } 1$$

$$= 0$$

$$RHD = Rf'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x} - 0}{x - 0}$$

$$= \lim_{h \rightarrow 0} \frac{(0+h)^2 \sin \frac{1}{0+h}}{0+h}$$

$$x = 0+h, h > 0$$

$$= \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h}}{h}$$

$$= \lim_{h \rightarrow 0} h \sin \frac{1}{h}$$

$$= 0 \times \text{a finite quantity b/w } -1 \text{ \& } 1$$

$$= 0$$

RHD & LHD exists and both are equal
 $\Rightarrow f'(0)$ exists and $f'(0) = 0$

Q2 A funcⁿ $f(x)$ is defined as follows:

$$f(x) = 1 + \sin x \quad \text{for } 0 < x < \frac{\pi}{2}$$

$$f(x) = 2 + \left(x - \frac{\pi}{2}\right)^2 \quad \text{for } x \geq \frac{\pi}{2}$$

Examine its derivability at $x = \frac{\pi}{2}$.

Soluⁿ

$$LHD = Lf'(0) = \lim_{x \rightarrow \frac{\pi}{2}-0} \frac{f(x) - f(\frac{\pi}{2})}{x - \frac{\pi}{2}}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}-0} \frac{[1 + \sin x] - 2}{x - \frac{\pi}{2}}$$

$$= \lim_{h \rightarrow 0} \frac{[1 + \sin(\frac{\pi}{2} - h)] - 2}{\frac{\pi}{2} - h - \frac{\pi}{2}}$$

$$= \lim_{x \rightarrow \frac{\pi}{2} - 0} \frac{\sin x - 1}{x - \frac{\pi}{2}}$$

$$= \lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{2} - h\right) - 1}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos h - 1}{-h} \quad \frac{0}{0}$$

$$= \lim_{h \rightarrow 0} \frac{\sin h}{1} \quad \text{L'Hop Hospital Rule}$$

$$= 0$$

$$\text{RHD} = \lim_{x \rightarrow \frac{\pi}{2} + 0} \frac{f(x) - f\left(\frac{\pi}{2}\right)}{x - \frac{\pi}{2}}$$

$$= \lim_{x \rightarrow \frac{\pi}{2} + 0} \frac{2 + \left(x - \frac{\pi}{2}\right)^2 - 2}{x - \frac{\pi}{2}}$$

$$= \lim_{x \rightarrow \frac{\pi}{2} + 0} \left(x - \frac{\pi}{2}\right)$$

$$= \lim_{h \rightarrow 0} \left[\frac{\pi}{2} + h - \frac{\pi}{2}\right]$$

$$= 0$$

$$\text{RHD} = \text{LHD} = 0$$

$$\Rightarrow f'\left(\frac{\pi}{2}\right) = 0$$

Q $f(x) = \begin{cases} x^2 - 1 & \text{when } x \geq 1 \\ 1 - x & \text{when } x < 1 \end{cases}$ at $x = 1$

$$f(1) = 1^2 - 1 = 0$$

Soln?

$$\text{L}f'(1) = \lim_{x \rightarrow 1 - 0} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1 - 0} \frac{(1 - x) - 0}{x - 1}$$

$$= \lim_{h \rightarrow 0} \frac{[1 - (1 + h)]}{1 + h - 1} = -1$$

$$\begin{aligned}
 \text{RHD} = Rf'(0) &= \lim_{x \rightarrow 1^+} \frac{f(x) - f(1)}{x - 1} \\
 &= \lim_{x \rightarrow 1^+} \frac{(x^2 - 1) - 0}{x - 1} \\
 &= \lim_{x \rightarrow 1^+} \frac{(x+1)(x-1)}{\cancel{x-1}} \\
 &= \lim_{h \rightarrow 0} \frac{[1+h+1]}{1} \\
 &= 2
 \end{aligned}$$

$\text{LHD} \neq \text{RHD} \Rightarrow$ Derivative at $x=1$ does not exist
 i.e. $f(x)$ is not derivable at $x=1$

Algebra of Derivatives

Let $f: [a, b] \rightarrow \mathbb{R}$ and $g: [a, b] \rightarrow \mathbb{R}$ be derivable at $c \in (a, b)$ then

- (i) λf is derivable at c and $(\lambda f)'(c) = \lambda f'(c)$ where $\lambda \in \mathbb{R}$
- (ii) $f+g$ is derivable at c and $(f+g)'(c) = f'(c) + g'(c)$
- (iii) $f \cdot g$ is derivable at c and $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$
- (iv) f/g is derivable at c and $(f/g)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{[g(c)]^2}$
 $g(c) \neq 0$
- (v) $\frac{1}{g}$ is derivable at c and $(\frac{1}{g})'(c) = -\frac{g'(c)}{[g(c)]^2}$, $g(c) \neq 0$

one of Roll's theorem is not true.

$f'(x)$ may be zero at a pt $c \in (a, b)$ without $f(x)$ satisfying all the three cond. of Roll's theorem

$$f(x) = \begin{cases} 0 & 0 \leq x \leq 1 \\ x+1 & 1 < x \leq 2 \end{cases}$$

not c'th at $x=1$

$$\text{LHL} = \lim_{x \rightarrow 1-0} 0 = 0$$

$$\text{RHL} = \lim_{x \rightarrow 1+0} f(x) = \lim_{h \rightarrow 0} (1+h+1)$$

$$= \lim_{h \rightarrow 0} [1+h+1] = 2$$

$\Rightarrow f$ is not c'th at $x=1$

$\Rightarrow f$ is not c'th on $[0, 2]$

$\Rightarrow f$ is not diff on $[0, 2]$

$$\text{Also } f(0) = 0$$

$$f(2) = 3$$

$$f(0) \neq f(2)$$

But $f'(x) = 0 \quad \forall x \in (0, 1) \subset (0, 2)$

$$\cancel{f'(0) = 0}$$

Rolle's Theorem:

(7)

If a funcⁿ $f: [a, b] \rightarrow \mathbb{R}$.

(i) f is c² on $[a, b]$

(ii) f diff. on (a, b)

(iii) $f(a) = f(b)$

then $\exists$ atleast one real no. $c \in (a, b)$ s.t. $f'(c) = 0$

